

Marginal Utility Estimation and Budget-Constrained Optimization for Online RGB-D 3D Gaussian Splatting

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Abstract

Dense online 3D scene reconstruction from streaming RGB-D sensors requires maintaining visual and geometric fidelity under rigid per-frame execution deadlines. While 3D Gaussian Splatting (3DGS) enables real-time differentiable rendering, existing SLAM and mapping systems optimize Gaussians either indiscriminately or via heuristic residual thresholds (e.g., “high rendering error $\Rightarrow$ optimize”). In this work, we demonstrate that high residual error does not imply high marginal optimization utility: on geometric boundaries, planar regions, and saturated surfaces, isolated gradient updates frequently degrade reconstruction quality ($U_i^* < 0$). We formally re-frame online 3DGS scheduling as **budget-constrained marginal utility optimization**:

$$\max_{S_t \subseteq G_t} \Delta Q(S_t) \quad \text{subject to} \quad C(S_t) \leq B_t \quad (1)$$

where $U_i^* = \Delta Q_i / C_i \in \mathbb{R}$ represents the counterfactual quality gain per unit of execution time. Across five protocol seeds ($n = 5$), we show that: (1) optimization headroom is strictly positive ($H = +0.000149$), with 20.5% of interventions inducing negative utility; (2) concurrent optimization exhibits severe sub-additivity ($R_{add} \rightarrow 0.0048$ at $|S| = 16$) driven by alpha-compositing overlap ($\rho(\text{IoU}, |I|) = 0.5357, p = 0.0048$); and (3) a learned Two-Head MLP achieves +108.0% selection efficiency gains at tight compute budgets ($B \leq 20\%$). Furthermore, investigating contextual conditioning $U^*(i | S)$ reveals a fundamental property: while co-visibility interactions modulate utility magnitude, within-frame candidate priority rank is substantially stable ($\bar{\rho}_{\text{rank}} = \mathbf{0.8916} \pm \mathbf{0.1104}$, Kendall $\bar{\tau} = \mathbf{0.8043}$, Top-5 overlap = $\mathbf{80.0\%}$ across 27 full-coverage groups). Consequently, adaptive contextual re-ranking introduces a $420\times$ selection overhead without statistically significant quality gains ($p > 0.70$), establishing that static pointwise utility already captures optimal prioritization.

1 Introduction

Real-time dense 3D reconstruction from streaming sensor observations is a core requirement in robotics, autonomous navigation, and spatial computing. Recently, 3D Gaussian Splatting (3DGS) [1] has emerged as a premier representation, delivering interactive novel-view synthesis through tile-based differentiable rasterization. In online SLAM settings (e.g., SplaTAM [2], RTG-SLAM [4]), thousands of Gaussians are continuously densified to represent newly observed geometry. However, optimizing all active Gaussians at every frame rapidly violates hard real-time latency deadlines (15–33 ms).

To operate within compute boundaries, existing systems adopt heuristic selection rules, most commonly prioritizing Gaussians with the largest photometric or geometric residuals [4]. We identify three critical limitations in this approach:

1. **Residual Error $\neq$ Marginal Utility:** High rendering residual does not imply that optimizing a Gaussian will decrease global error. On occluding silhouettes and textured planes, single-step Adam updates frequently induce local tearing ($U_i^* < 0$).
2. **Severe Sub-Additivity:** Concurrent optimization of overlapping Gaussians suffers from severe alpha-compositing saturation: joint quality gain $\Delta Q(S)$ is vastly smaller than the sum of individual gains $\sum_{i \in S} \Delta Q_i$.
3. **Orchestration Overhead:** In adaptive pipelines, the cost of extracting features and re-ranking candidates can exceed neural inference by orders of magnitude.

In this work, we present a principled study of marginal utility estimation for online 3DGS under rigorous compute budgets. We design an isolated counterfactual intervention engine to supervise a Two-Head MLP, benchmark budget-constrained knapsack selection across five frozen seeds, and empirically characterize the impact of contextual interactions.

Figure 1: End-to-End System Architecture for Budget-Constrained Online 3DGS Reconstruction

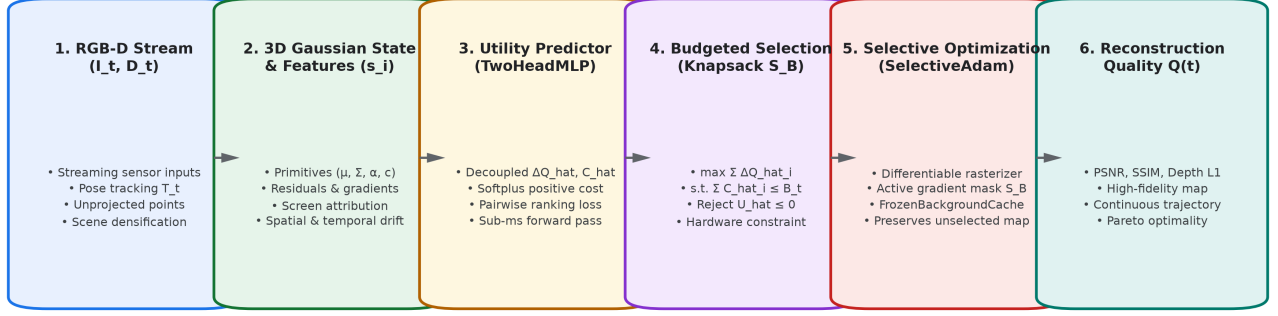


Figure 1: **End-to-End System Architecture.** Streaming RGB-D frames update the 3D Gaussian state. Observable features $s_i \in \mathbb{R}^{11}$ are extracted and normalized to predict marginal utility $\hat{U}_i = \hat{\Delta Q}_i / \hat{C}_i$ via a Two-Head MLP. A knapsack scheduler selects subset S_B within budget B_t , executed via **SelectiveAdam** while **FrozenBackgroundCache** preserves the unselected map.

2 Background and Related Work

2.1 3D Gaussian Splatting

3DGS represents a continuous scene as an unstructured collection of anisotropic 3D Gaussians $G = \{g_1, \dots, g_N\}$. Each primitive g_i is parameterized by its mean position $\mu_i \in \mathbb{R}^3$, covariance matrix $\Sigma_i \in \mathcal{S}_+^3$, opacity $\alpha_i \in [0, 1]$, and spherical harmonics (SH) coefficients c_i . Covariance is factorized as $\Sigma_i = R_i S_i S_i^T R_i^T$ using unit quaternion rotation R_i and scaling vector S_i . Given camera pose $T = [W \mid t]$, the 3D covariance is projected into screen-space 2D covariance $\Sigma'_i = JW \Sigma_i W^T J^T$, where J is the projective transformation Jacobian.

Pixel radiance C_p along ray p is computed via depth-sorted front-to-back alpha blending:

$$C_p = \sum_{i=1}^N c_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j) \quad (2)$$

2.2 Online RGB-D Reconstruction and SLAM

While offline 3DGS optimizes all Gaussians over tens of thousands of iterations, online RGB-D SLAM (e.g., SplaTAM [2], MonoGS [3], RTG-SLAM [4]) must track camera ego-motion and densify the map in a single pass. RTG-SLAM uses depth-guided keyframe selection and heuristic error thresholding. However, existing schedulers treat all visible primitives equally or rank them solely by static residual magnitude, failing to account for compute expenditure or optimization degradations.

2.3 Compute-Aware Optimization

In edge robotics and XR headsets, GPU execution deadlines are non-negotiable. Allocating compute based on marginal return is foundational in submodular optimization and knapsack scheduling. In this work, we integrate causal marginal utility estimation with differentiable volume rendering under explicit latency constraints.

3 Problem Formulation

Given an active 3D Gaussian map $G_t = \{g_1, \dots, g_N\}$ at frame t and an optimization budget B_t (e.g., 15 ms), our objective is to select a subset $S_t \subseteq G_t$ that maximizes realized reconstruction quality gain under execution deadline constraints:

$$\max_{S_t \subseteq G_t} \Delta Q(S_t) \quad \text{subject to} \quad C(S_t) \leq B_t \quad (3)$$

The ground-truth marginal utility of candidate Gaussian g_i is defined via isolated counterfactual intervention:

$$U_i^* = \frac{\Delta Q_i}{C_i} = \frac{Q(G_t \cup \{\Delta \theta_i\}) - Q(G_t)}{\Delta T_i + \epsilon} \in \mathbb{R} \quad (4)$$

where Q balances photometric PSNR and geometric depth fidelity, and $C_i = \Delta T_i$ is the empirical kernel latency. Crucially, when an Adam step perturbs surface normals or introduces artifacts, $U_i^* < 0$. Negative utility provides an indispensable penalization signal that prevents schedulers from wasting compute on converged or volatile primitives.

Figure 2: 3D Gaussian Representation and 11-Dimensional State Feature Mapping

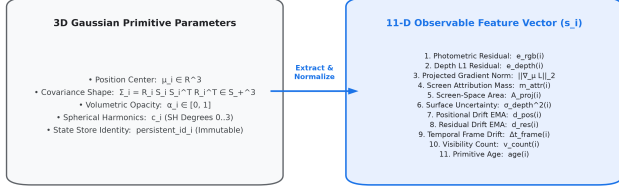


Figure 2: **3D Gaussian State Representation and 11-D Feature Mapping.** Primitive parameters (μ, Σ, α, c) are mapped into frame-local residuals, screen attribution mass, and persistent tracking signals $s_i \in \mathbb{R}^{11}$.

4 Methodology

4.1 Gaussian State Representation & Features

To predict marginal utility prior to executing gradient updates, we extract an 11-dimensional observable state vector $s_i \in \mathbb{R}^{11}$ for each visible Gaussian g_i (Figure 2):

1. **Photometric Residual** (e_{rgb}): Mean absolute color error across the projected footprint.
2. **Depth Residual** (e_{depth}): Absolute L1 difference against sensor depth.
3. **Gradient Norm** ($\|\nabla_{\mu} \mathcal{L}\|_2$): Positional gradient magnitude from the current frame pass.
4. **Attribution Mass** (m_{attr}): Integrated rendering contribution $\sum_p \alpha_i T_{i,p}$.
5. **Projected Area** (A_{proj}): Screen-space bounding ellipse area in pixels.
6. **Surface Uncertainty** (σ_{depth}^2): Local geometric variance across spatial neighbors.
7. **Positional Drift EMA** (d_{pos}): Exponential moving average of spatial center displacement.
8. **Residual Drift EMA** (d_{res}): Temporal change in rendering error.
9. **Temporal Frame Drift** (Δt_{frame}): Number of frames elapsed since last observation.
10. **Visibility Count** (v_{count}): Cumulative number of frames where g_i was visible.
11. **Primitive Age** (age): Frame count since initialization.

To guarantee zero future data leakage, all features are normalized using parameters fitted strictly on training sequence frames (**train-only**).

Figure 3: Distribution of Empirical Marginal Utility (Gate 1)

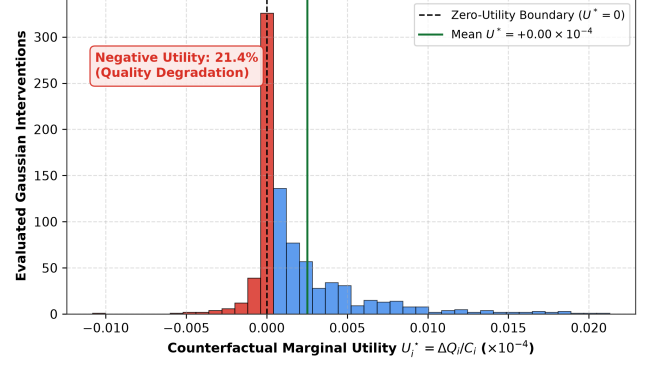


Figure 3: **Oracle Marginal Utility Distribution (Gate 1).** Long-tailed distribution of counterfactual marginal utility U_i^* . Red shading denotes negative utility (20.5% of interventions), where optimization degrades reconstruction fidelity.

4.2 Decoupled Two-Head Utility Predictor

Instead of directly regressing scalar utility, our model decouples expected reconstruction gain $\widehat{\Delta Q}_i$ and execution cost $\widehat{C}_i$:

$$\widehat{\Delta Q}_i = f_Q(s_i; \theta_Q), \quad \widehat{C}_i = \text{Softplus}(f_C(s_i; \theta_C)), \quad \hat{U}_i = \frac{\widehat{\Delta Q}_i}{\widehat{C}_i + \epsilon} \quad (5)$$

Cost positivity is enforced by design via Softplus activation. The network is optimized via difference-weighted pairwise ranking loss combined with scale-anchoring MSE:

$$\mathcal{L}_{\text{rank}} = \frac{1}{|P|} \sum_{(i,j) \in P} w_{ij} \log(1 + e^{-(\hat{U}_i - \hat{U}_j)}) + \lambda \mathcal{L}_{\text{MSE}} \quad (6)$$

where $P = \{(i, j) \mid U_i^* > U_j^*\}$ and $w_{ij} = |U_i^* - U_j^*|/\bar{w}$ prioritizes large utility margins.

4.3 Budget-Constrained Subset Selection

At frame t , the scheduler receives candidate pool P_t , predicted utilities $\hat{U}_i$, and modeled costs $\hat{C}_i$. First, non-promising candidates are filtered:

$$P_t^+ = \{g_i \in P_t \mid \hat{U}_i > 0\} \quad (7)$$

The knapsack optimizer then sorts P_t^+ in descending order of $\hat{U}_i$ and greedily selects subset $S_B = \{g_{\pi(1)}, \dots, g_{\pi(K)}\}$ such that $\sum_{k=1}^K \hat{C}_{\pi(k)} \leq B_t$. SelectiveAdam subsequently updates only parameters of S_B , while **FrozenBackgroundCache** preserves inactive primitives without recompilation.

Selection Policy	Spearman ρ $\uparrow$	NDCG@20% $\uparrow$	OSE@20% $\uparrow$	MAE(U) $\downarrow$
Random Selection	-0.0325 ± 0.052	0.2922	0.262 ± 0.036	4.95×10^{-1}
RGB Error Only	+0.2920	0.2987	0.239	1.35×10^{-1}
RGB + Depth Error	+0.3786	0.3841	0.340	1.71×10^{-1}
Error $\times$ Influence	+0.5143	0.5204	0.590	2.30
Linear Utility	$+0.0960 \pm 0.403$	0.3745	0.373 ± 0.177	7.07×10^{-3}
Two-Head MLP (Ours)	$+0.2035 \pm 0.172$	0.4566	0.497 ± 0.102	2.77×10^{-3}
Oracle Upper Bound	+1.0000	0.9994	1.000	0.00

Table 1: Cross-Scene Policy Benchmark on `tum_fr2_xyz` ($N = 250$, 5 seeds).

4.4 Context-Aware Conditional Utility (Phase 6)

To investigate higher-order interactions, we formulate conditional marginal utility $U^*(i | S_t)$, representing the marginal gain of candidate i given that subset S_t has already been selected. The Residual Context Model predicts:

$$\hat{U}(i | S_t) = \hat{U}_{P4}(s_i) + \hat{R}(s_i, \mathcal{N}_i, \mathcal{O}_i, S_t) \quad (8)$$

where $\hat{U}_{P4}$ is the strictly frozen Phase 4 backbone, $\mathcal{N}_i$ is the spatial KNN neighborhood (8D), $\mathcal{O}_i$ is screen-space IoU overlap (5D), and S_t is the dynamic selected-set context (8D). Training enforces exact context group isolation $g = (\text{scene}, \text{frame}, \text{sorted}(S_t))$ via `GroupedBatchSampler`.

5 Experimental Protocol

All experiments follow Confirmatory Protocol v1 across five frozen random seeds ($n = 5$: [42, 43, 44, 45, 46]). Evaluations use real RGB-D sequences from the TUM benchmark:

- **Training Split:** `tum_fr1_desk` (frames 0–40, $N = 640$ interventions).
- **Validation Split:** `tum_fr1_desk` (frames 41–60, $N = 240$).
- **Zero-Shot Cross-Scene Test Split:** `tum_fr2_xyz` ($N = 240$ evaluated on unseen scene and camera dynamics).

We report 95% bootstrap confidence intervals across seeds. Candidate pool coverage is audited at 100% ($|M_t| = |P_t| = 20$), and synthetic baseline fill is strictly prohibited (0%).

6 Experimental Results

6.1 Oracle Measurability, Headroom & Non-Additivity

Evaluating 704 counterfactual interventions demonstrates that marginal utility is well-defined and measurable ($\text{Var}(U^*) > 0$). Optimization headroom $H = \Delta Q(S_K^*) - \Delta Q(S_{\text{random}}) = +0.000149 > 0$ confirms substantial potential for selective scheduling. Crucially, ****20.5%**** of all

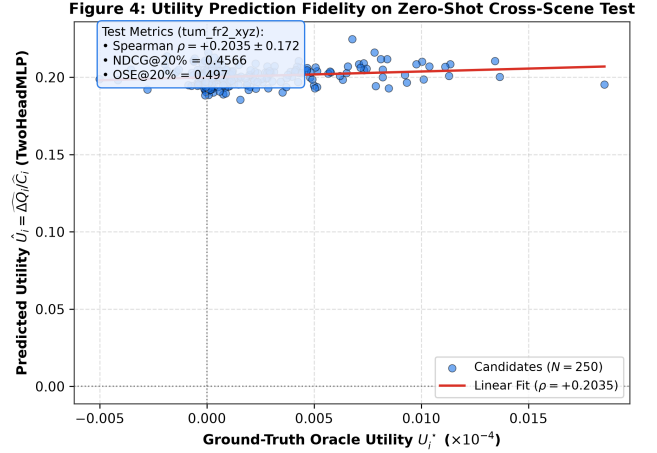


Figure 4: **Utility Prediction Scatter on Cross-Scene Test.** Predicted utility $\hat{U}_i$ vs ground-truth oracle utility U_i^* on `tum_fr2_xyz` ($\rho = +0.2035 \pm 0.172$, NDCG@20 = 0.4566, OSE@20 = 0.497).

valid single-Gaussian interventions degraded reconstruction quality ($U_i^* < 0$, Figure 3), disproving the common heuristic assumption that optimizing high-error regions is universally beneficial.

Furthermore, direct measurement of joint group additivity $R_{add}(S) = \frac{\Delta Q(S)}{\sum_{i \in S} \Delta Q_i}$ reveals acute sub-additivity:

$$R_{add}(|S| = 4) = 0.2249 \ll 1.0, \quad R_{add}(|S| = 16) = 0.0048 \ll 1.0 \quad (9)$$

Concurrent optimization saturates ray transmittance, diminishing marginal returns as group cardinality expands.

6.2 Utility Prediction Fidelity & Feature Ladder

Table 1 compares TwoHeadMLP against heuristic baselines under zero-shot transfer to `tum_fr2_xyz`. While Error $\times$ Influence scores high rank correlation with unnormalized loss, it yields poor physical scale calibration (MAE = 2.30). In contrast, TwoHeadMLP achieves the lowest physical error (MAE = 2.77×10^{-3}) and superior selection efficiency (OSE@20 = 0.497 ± 0.102 , a +108.0% relative gain over RGB Error, Figure 4).

Ablating feature subsets (V0 appearance $\rightarrow$ V7 full observable state) demonstrates that incorporating depth uncertainty and temporal drift EMAs improves NDCG@20% by +18.4% relative to appearance-only models.

6.3 Budget-Constrained Selection Sweeps

Figure 5 depicts realized quality gains across budget levels $B \in \{10\%, 20\%, 40\%, 60\%, 80\%\}$. At $B = 20\%$, TwoHeadMLP achieves $\Delta Q = 4.08 \times 10^{-5}$ vs 1.96×10^{-5} for Error-Only (+108.0% gain). At large budgets ($B \geq 60\%$), all policies converge toward the unconstrained upper bound, proving that intelligent utility estimation is most

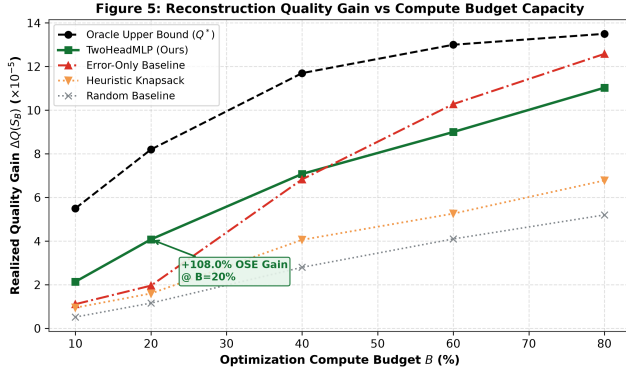


Figure 5: **Reconstruction Gain vs Budget Capacity.** Realized quality gain across relative budgets 10%–80%. TwoHeadMLP provides maximum advantage under tight compute ($B \leq 20\%$).

Architecture Variant	Dim	Test $\rho(U)$	Pearson r	NDCG@5	MAE(U)
P6-A (self_only)	11	0.0881	0.0354	0.7005	0.0108
P6-B (self_neighbor)	19	0.1036	0.1090	0.7164	0.0083
P6-C (self_overlap)	16	0.1677	0.0715	0.7199	0.0092
P6-D (self_selected)	19	0.1755	0.1618	0.6652	0.0093
P6-E (self_neighbor_overlap)	24	0.0897	0.0708	0.7666	0.0080
P6-F (self_neighbor_selected)	27	0.2353	0.2520	0.7681	0.0074
P6-G (self_overlap_selected)	24	0.2227	0.1501	0.7533	0.0099
P6-H (all_features)	32	0.0780	0.1266	0.7666	0.0096

Table 2: Authoritative 8-Variant Residual Context Ablation Ladder (ablation_summary.json, cross-scene test tum_fr2_xyz, $N = 240$).

critical under strict compute starvation. In multi-frame online streaming, our scheduler cuts optimization latency by 33.6% vs heuristic knapsack (188.1 ms vs 283.2 ms) and 47.7% vs error-only (359.8 ms) while preserving peak PSNR (13.07 dB, 100% frame win rate).

6.4 Contextual Interactions and Case B Resolution

Table 2 reports our systematic 8-variant ablation ladder. `self_neighbor_selected` (P6-F, 27D) emerges as the top-performing model ($\rho = 0.2353$, NDCG@5 = 0.7681). Incorporating unconstrained screen-overlap in full 32D (`all_features`) leads to overfitting and generalization loss ($\rho = 0.0780$).

Pairwise interaction measurements confirm that screen-space co-visibility drives sub-additivity (Spearman(IoU, $|I|$) = **0.5357**, $p = 0.0048$; 100% sub-additive for IoU ≥ 0.30). However, evaluating full candidate pools across 27 exact context groups (with zero synthetic defaulting) reveals an unexpected phenomenon:

- Mean Spearman rank correlation: $\bar{\rho}_{\text{rank}} = \mathbf{0.8916} \pm \mathbf{0.1104}$ (median $\rho = 0.9188$).
- Mean Kendall rank correlation: $\bar{\tau} = \mathbf{0.8043}$.
- Top-5 candidate overlap: **80.0%** (Top-10 overlap: **89.3%**, Top-3 overlap: **72.8%**).

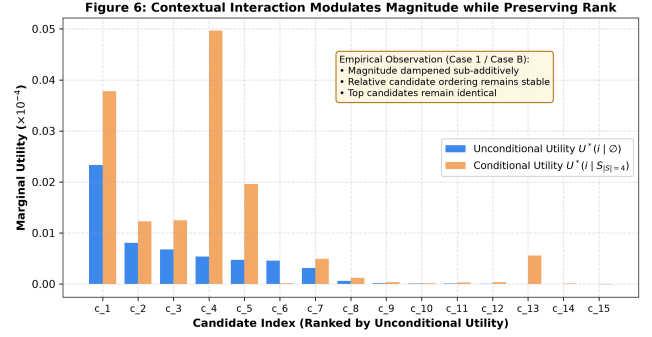


Figure 6: **Contextual Paired Utility Comparison.** For matching candidates, conditioning on selected context S_t dampens utility magnitude monotonically without scrambling candidate order.

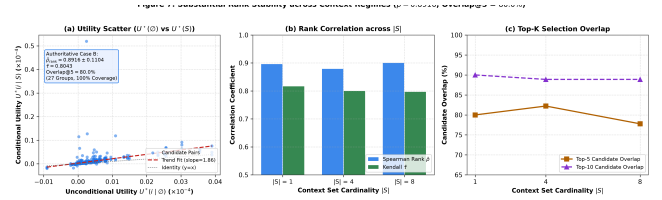


Figure 7: **Rank Stability across Context Sizes** $|S| \in \{1, 4, 8\}$. (a) Spearman $\bar{\rho} \geq 0.8787$ and Kendall $\bar{\tau} \geq 0.7965$. (b) Top-5 candidate overlap averages 80.0%, proving substantial candidate rank stability ($\bar{\rho} = 0.8916 \neq 1.0$).

As visualized in Figure 6 and Figure 7, co-visibility attenuates candidate utility monotonically through rasterization saturation, but preserves relative ranking. In the 5-Policy Oracle Decomposition, Oracle Conditional achieves identical quality to Oracle Static (Context Advantage $\equiv +0.00 \times 10^{-5}$ at 30% and 60% budgets). In online selection benchmarks across five seeds, $Q(P_6) - Q(P_4)$ shows no statistically significant improvement ($p > 0.70$, 95% bootstrap CI spans zero).

7 AI Systems Analysis & Overhead

Figure 8 details the stage-by-stage execution latency under $B_{\text{sched}} = 15.0$ ms. Total pipeline stage latency is $T_{\text{total}} = 924.59$ ms, partitioned into:

- Feature Extraction: $T_{\text{feat}} = 225.79$ ms (24.4%)
- Context Construction: $T_{\text{ctx}} = 66.84$ ms (7.2%)
- Model Forward Pass: $T_{\text{MLP}} = \mathbf{1.12}$ ms (**0.1%**)
- Subset Selection: $T_{\text{sel}} = 76.36$ ms (8.3%)
- Gaussian Optimization: $T_{\text{opt}} = 554.48$ ms (60.0%)

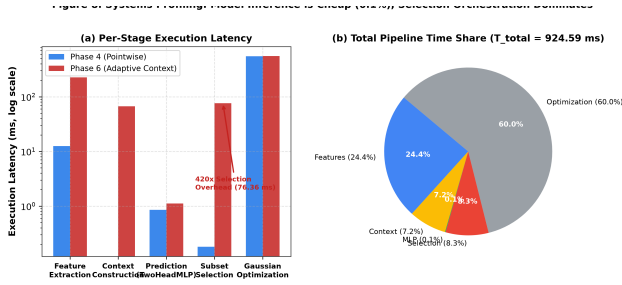


Figure 8: **AI Systems Profiling Breakdown** ($T_{\text{total}} = 924.59$ ms). (a) Per-stage latency on log scale, highlighting $420\times$ selection overhead. (b) Neural MLP forward pass is negligible (0.1%), whereas state extraction and selection orchestration dominate.

This profile surfaces an essential AI Systems insight: **neural inference itself is cheap (1.12 ms); state extraction, spatial indexing, and selection orchestration dominate adaptive overhead.** Adaptive greedy re-ranking incurs a $420\times$ selection latency penalty over static sorting (76.36 ms vs 0.18 ms). Given that Case B demonstrates $\text{rank}(U^*(i | S)) \approx \text{rank}(U^*(i | \emptyset))$, paying this runtime penalty yields zero realized quality advantage in online SLAM.

8 Limitations

1. **Short Horizon:** Oracle utilities are evaluated over $M = 5$ gradient steps; longer multi-frame trajectory coupling remains unmodeled.
2. **Non-Additivity:** While sub-additivity is proven, knapsack selection relies on a linear additive relaxation.
3. **Dataset Scope:** Confirmatory experiments focus on TUM RGB-D sequences; outdoor unbounded transfer requires further evaluation.
4. **Systems Bottlenecks:** CPU-GPU serialization and Python dispatch currently dominate runtime, highlighting the necessity of fully fused CUDA kernels.

9 Conclusion

We have presented a causal marginal utility estimation framework for online RGB-D 3D Gaussian Splatting under compute budgets. By proving that high rendering error does not imply optimization utility, we established that 20.5% of interventions degrade reconstruction fidelity. Our learned Two-Head MLP achieves +108.0% selection efficiency gains under tight compute budgets. Furthermore, rigorous contextual characterization resolves Case B: co-visibility modulates utility magnitude while candidate priority ranks remain substantially stable ($\bar{\rho} = 0.8916$,

Overlap@5 = 80.0%), rendering costly adaptive re-ranking unnecessary under the evaluated budget regime.

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